

Nguyen Minh Khoa

AI Engineer & Researcher

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I'm Khoa Nguyen, an AI engineer and Research Assistant at AiTA Lab, FPT University.

My recent work includes handwritten mathematical expression recognition and text-to-molecule generation. I have worked across data preparation, model experiments, and inference integration, with a preference for research that can be checked in working software.

I'm now focusing on AI for manufacturing, especially visual systems that must cope with limited data and changing operating conditions. I'm looking for work that gives equal weight to careful experiments and hands-on implementation.

I'm open to AI engineering, research engineering, and applied machine-learning collaborations.

RESEARCH EXPERIENCE

Research Assistant | AiTA Lab, FPT University | Oct 2024 - Present

- Co-authored two HMER papers at ICDAR and in the Journal of Information and Telecommunication, contributing masked-language pretraining and multi-task learning experiments.
- First-authored LexiChem (APWeb-WAIM 2026): developed methods, ran ablations and presented the work in Da Nang. Co-first author of ChemAligner-T5 (ICCIES 2026).
- Co-authored CorrTie (APWeb-WAIM 2026), contributing critical review and methodological refinement.

SELECTED PROJECTS

LexiChem | AI & Inference Contributor | 2025 - 2026

- Built the AI inference workflow so users can generate molecules from text, validate structures and revisit saved runs using FastAPI, Triton, RDKit and MongoDB.
- Built a three-dataset training pipeline and ran ablations for the first-author APWeb-WAIM paper.

NexOps | AI, Simulation & Platform Lead | 2025 - Present

- Built the local MES/SCADA runtime connecting simulated machines to production planning, alarms and OEE dashboards through MQTT, FastAPI and TimescaleDB.
- Connected a Bambu Lab printer over LAN, demonstrating real device control in the local prototype.

SELECTED PUBLICATIONS

LexiChem: Shared-Latent Alignment for Faithful Text-to-Molecule Generation in SELFIES Space

APWeb-WAIM 2026 | First author | Accepted & presented

Mask CoMER: Enhancing Handwritten Mathematical Expression Recognition with Masked Language Pretraining and Regularization

ICDAR 2025 | Co-first author | Published; proceedings 2026

Enhancing handwritten mathematical expression recognition with convolutional block attention refinement module and multi-task learning

Journal of Information and Telecommunication | Co-author | Published online 19 August 2026

EDUCATION

Bachelor of Science in Artificial Intelligence | FPT University, HCMC

2022 - 2026 | Academic requirements completed; degree conferral expected November 2026.

AWARDS & SCHOLARSHIPS

50% tuition scholarship, FPT University (2022) | **RESFES Second Prize**, 2025 & 2026.

TrendRadar team awards/funding: Demo Pitching Day, VND 50M; Innovation Quest, VND 40M (2025).

TECHNICAL SKILLS

AI & research: Python, PyTorch, Transformers, model experiments and ablations, HMER, cross-modal learning, text-to-molecule generation

Systems & data: FastAPI, NVIDIA Triton, RDKit, PostgreSQL/TimescaleDB, MongoDB, MQTT, Docker, Git

SELECTED CERTIFICATIONS

AIO | AI VIET NAM: Machine Learning; Deep Learning; Computer Vision & NLP; GenAI & LLMs.

Machine Learning (Stanford Online / DeepLearning.AI, 2023) | Deep Learning (DeepLearning.AI, 2024)